Book of Proof: Condensed Version with All Exercise Solutions

KS

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1 Fundamentals

1.1 Introduction to Sets

All of mathematics can be described with sets. Set theory is *a* perfectly suitable language to express and describe all types of mathematical structures.

Think of a *set* a collection of things. The things are called *elements* of the set. It's somewhat worth noting here that we're only concerned with sets that express mathematical entities like functions, numbers, points, etc. Set construction: $\{2, 3, 4, 5\}$ is a set which has 4 elements. Some sets have infinitely many elements, *not all* sets are finite.

We say that two sets are *equal* if they contain exactly the same elements. So: $\{2, 3, 4, 5\} = \{2, 4, 3, 5\}$ but $\{2, 3, 4, 5\} \neq \{2, 4, 6, 8\}$.

To express that an element is *member* of a set, we write $2 \in A$, where A is a set containing 2 as an element. Some sets are very important and we reserve special symbols for them, like:

$$\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}, \mathbb{C}$$

Sets don't have to contain only numbers, for example $B = \{T, F\}$ or $C = \{a, b, c, d, e\}$, or $D = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$. A set can have *other sets as elements*:

$$K = \{1, \{2, 3\}, \{2, 4\}\}$$

It's really important to note here that K has 3 elements. $1, \{2, 3\}, \{2, 4\} \in K$ but $2, 3, 4 \notin K$.

If a set X is finite, we say that X's *cardinality* is the number of elements it has, and this number is denoted by $|X|$. An example of a special set is $\varnothing$. It is called the empty set and it enjoys the unique status of having no members and thus $|\varnothing| = 0$.

We have special *set builder notation* to describe sets that are too large or too complex to list between braces. Consider the infinite set of even integers $E = \{\ldots, -4, -2, 2, 4, 6 \ldots\}$. We can instead write

$$E = \{2n | n \in \mathbb{Z}\}.$$

We read it "the set of all things of form" $2n$, | "such that", n "is a member of" the set of integers. The following generalised structure follows for set-builder notation:

$$X = \{\text{expression} : \text{rule}\};$$

where the elements of X are understood to be *all values of expression* that are specified by the *rule*.

Example 1.1.1.

1. $\{n : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17...\}$
2. $\{n \in \mathbb{N} : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17\}$
3. $\{x \in \mathbb{R} : x^2 - 2 = 0\} = \{\sqrt{2}, \sqrt{-2}\}$
4. $\{x \in \mathbb{Z} : x^2 - 2 = 0\} = \varnothing$
5. $\{x \in \mathbb{Z} : |2x| < 4\} = \{-1, 0, 1\}$

Example 1.1.2. Describe the set $A = \{7a + 3b : a, b \in \mathbb{Z}\}$.

Solution. The set A is equivalent to the expression $7a + 3b$, where the rule for finding a, b is that they are members of the set of integers. Consider which integers. Let n be any integer, then $n = 7n+3(-2n)$, such that $n = a$ and $b = -2n$. So, $n \in A$. It follows that A contains only integers, and furthermore, *every integer is an element of* A. So $A = \mathbb{Z}$. □

Important: Set-builder notation for $\mathbb{Q}$:

$$\mathbb{Q} = \{x : x = \frac{m}{n} \text{ where } m, n \in \mathbb{Z} \wedge n \neq 0\}$$

1.2 Exercises from Section 1

Exercise 1.1. Write each of the following sets by listing their elements between braces.

$1.\{5x - 1 : x \in \mathbb{Z}\}$ $9.\{x \in \mathbb{R} : \sin \pi x = 0\}$

$2.\{3x + 2 : x \in \mathbb{Z}\}$ $10.\{x \in \mathbb{R} : \cos x = 1\}$

$3.\{x \in \mathbb{Z} : -2 \leq x \leq 7\}$ $11.\{x \in \mathbb{Z} : |x| < 5\}$

$4.\{x \in \mathbb{N} : -2 < x \leq 7\}$ $12.\{x \in \mathbb{Z} : |2x| < 5\}$

$5.\{x \in \mathbb{R} : x^2 = 3\}$ $13.\{x \in \mathbb{Z} : |6x| < 5\}$

$6.\{x \in \mathbb{R} : x^2 = 9\}$ $14.\{5x : x \in \mathbb{Z}, |2x| \leq 8\}$

$7.\{x \in \mathbb{R} : x^2 + 5x = -6\}$ $15.\{5a + 2b : a, b \in \mathbb{Z}\}$

$8.\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\}$ $16.\{6a + 2b : a, b \in \mathbb{Z}\}$.

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Exercise 1.2. Write each of the following sets in set-builder notation.

$17.\{2, 4, 8, 16, 32, 64, \dots\}$ $23.\{3, 4, 5, 6, 7, 8\}$

$18.\{0, 4, 16, 36, 64, 100, \dots\}$ $24.\{-4, -3, -2, -1, 0, 1, 2\}$

$19.\{\dots, -6, -3, 0, 3, 6, 9, 12, 15\}$ $25.\{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \dots\}$

$20.\{\dots, -8, -3, 2, 7, 12, 17, \dots\}$ $26.\{\dots, \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, \dots\}$

$21.\{0, 1, 4, 9, 16, 25, 36, \dots\}$ $27.\{\dots, -\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \frac{5\pi}{2}, \dots\}$

$22.\{3, 6, 11, 18, 27, 38, \dots\}$ $28.\{\dots, -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, \dots\}$.

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Exercise 1.3. Find the following cardinalities.

$29.|\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}|$ $34.|\{x \in \mathbb{N} : |x| < 10\}|$

$30.|\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}|$ $35.|\{x \in \mathbb{Z} : x^2 < 10\}|$

$31.|\{\{\{1\}, \{2, \{3, 4\}\}, \varnothing\}\}|$ $36.|\{x \in \mathbb{N} : x^2 < 10\}|$

$32.|\{\{\{1, 4\}, a, b, \{\{3, 4\}\}, \{\varnothing\}\}\}|$ $37.|\{x \in \mathbb{N} : x^2 < 0\}|$

$33.|\{x \in \mathbb{Z} : |x| < 10\}|$ $38.|\{x \in \mathbb{N} : 5x \leq 20\}|$.

Exercise 1.4. Sketch the following sets of points in the xy-plane.

$$39.\{(x,y) : x \in [1,2], y \in [1,2]\} \qquad 46.\{(x,y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 1\}$$
$$40.\{(x,y) : x \in [0,1], y \in [1,2]\} \qquad 47.\{(x,y) : x, y \in \mathbb{R}, y \geq x^2 - 1\}$$
$$41.\{(x,y) : x \in [-1,1], y = 1\} \qquad 48.\{(x,y) : x, y \in \mathbb{R}, x > 1\}$$
$$42.\{(x,y) : x = 2, y \in [0,1]\} \qquad 49.\{(x, x+y) : x \in \mathbb{R}, y \in \mathbb{Z}\}$$
$$43.\{(x,y) : |x| = 2, y \in [0,1]\} \qquad 50.\{(x, \frac{x^2}{y}) : x \in \mathbb{R}, y \in \mathbb{N}\}$$
$$44.\{(x,x^2) : x \in \mathbb{R}\} \qquad 51.\{(x,y) \in \mathbb{R}^2 : (y-x)(y+x) = 0\}$$
$$45.\{(x,y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\} \qquad 52.\{(x,y) \in \mathbb{R}^2 : (y-x^2)(y+x^2) = 0\}.$$

1.3 The Cartesian Product

Think of the Cartesian product as a type of multiplication.

$$A \times B$$

In order to understand products better we will define an *ordered pair*.

Definition 1.3.1. An ordered pair is a list of the form x, y of two things, namely x and y, enclosed in parenthesis and separated by a comma.[1]

There are many ordered pairs, consider the ordered pair $(2,4)$. Now, if we swap the numbers around, that would be an ordered pair as well, i.e. $(4,2)$. However, these two pairs are not identical, meaning

$$(2,4) \neq (4,2).$$

Ordered pairs can be used to describe points on a plane (as it is done in Calculus and Linear Algebra for example). Similarly to our previous talk about sets, ordered pairs need not be *just* numbers, they can be letters as well.

Example 1.3.1. The set $(\{2,3\},\{3,4\})$ can be an ordered pair of two sets, say A, B, then $(\{2,3\},\{3,4\}) = (A,B)$;
α, β can also be a pair; finally, $((2,3),(3,4))$ can be an ordered pair of an ordered pair.

So, the following definition of Cartesian Products is important:

Definition 1.3.2. The Cartesian Product of two sets A, B is a set, denoted by

$$A \times B$$

and defined as

$$A \times B = \{(a,b) : a \in A, b \in B\}.$$

The following is a more visual example of how ordered pairs are formed.

[1] Additionally, one can use the Kuratowski ordered pair definition, i.e. $\{\{x\},\{x,y\}\} = (x,y)$. However, the one given above, even though slightly cryptic is easier to grasp for someone who has never encountered sets before.

Example 1.3.2. Let $A = \{k, l, m\}, B = \{o, p\}$ for A, B some sets, then the product of A with B is

$$A \times B = \{(k, o), (k, p), (l, o), (l, p), (m, o), (m, p)\}.$$

Schematically, one can think of these pairs in terms of rows and columns or in terms of a corrdinate system where A is the x-axis and B is the y-axis.

The following is a fact:

Fact 1.3.1. If A, B are finite sets, then $|A \times B| = |A| \cdot |B|$.

Example 1.3.3. Let $A = \{1, 2, 3, 4, 5, 6\}$ and suppose these are dice values. The Cartesian Product of A with itself is

$$A \times A = A^2.$$

The product of A with itself will be a finite set of ordered pairs.

$$|A \times A| = |A| \cdot |A| = 6 \cdot 6 = 36.$$

Think of the product as rolling a dice twice an recording the outcome, where the first roll corresponds to the first result and the second to the second result.

Example 1.3.4. Another few very useful examples are the sets:

$$\begin{aligned} \mathbb{R} \times \mathbb{R} &= \{(x, y) : x, y \in \mathbb{R}\} \\ \mathbb{R} \times \mathbb{N} &= \{(x, y) : x \in \mathbb{R}, y \in \mathbb{N}\} \\ \mathbb{N}^2 &= \{(x, y) : x, y \in \mathbb{N}\}. \end{aligned}$$

The first example is best described as a coordinate system or an xy-plane. The second example is simply all points on the plane, whose second coordinate is a natural number. The third example, is best described as the first quadrant of the xy-plane, where every x, y are natural numbers.

It is possible for one factor of a Cartesian product to be a Cartesian Product itself. Suppose the following:

$$\mathbb{R} \times (\mathbb{N} \times \mathbb{Z}) = \{x, (y, z) : x \in \mathbb{R}, (y, z) \in \mathbb{N} \times \mathbb{Z}\}.$$

There is a different way to think about these pairs. They are called *ordered triples*. The brackets are optional in the above example, one can also write

$$\mathbb{R} \times \mathbb{N} \times Z = \{(x, y, z) : x \in \mathbb{R}, y \in \mathbb{N}, z \in Z\}.$$

In general we can define

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_1 \in A_1, x_2 \in A_2, \ldots, x_n \in A_n\},$$

or more succinctly,

$$A_1 \times A_2 \times \cdots \times A_n = \{(x_1, x_2, \ldots, x_n) : x_i \in A_i\} \text{ for } i = \{1, 2, 3, \ldots, n\}$$

Example 1.3.5. Represent the two sides of a coin by the set $S = (H, T)$, the possible outcomes of tossing the coin seven times in a row can be described as:

Solution. This is the representation

$$S^7 = \{(x_1, x_2, \ldots, x_7) : x_1, x_2, \ldots, x_7 \in S\}.$$

Now to find the possible outcomes:

$$|S^7| = 7 \cdot 7 \cdot \ldots 7 = 128.$$

□

I believe the following to be clear but still worth noting. Notice that the cardinality of the product of S with itself seven times is written inside the || and not outside. This might seem like a silly thing to say, however syntax is very important.

1.4 Exercises from Section 2

Exercise 1.5. Write out the indicated sets by listing their elements between braces.

1. Suppose $A = \{1, 2, 3, 4\}, B = \{a, c\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A \times A$
   $(d) B \times B$
   $(e) \varnothing \times B$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) B^3$

2. Suppose $A = \{pi, e, 0\}, B = \{0, 1\}$.

   $(a) A \times B$
   $(b) B \times A$
   $(c) A^2$
   $(d) B^2$
   $(e) A \times \varnothing$
   $(f) (A \times B) \times B$
   $(g) A \times (B \times B)$
   $(h) A \times B \times B.$

3. $\{x \in \mathbb{R} : x^2 = 2\} \times \{a, c, e\}$.

4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\}$.

5. $\{x \in \mathbb{R} : x^2 = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$.

6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\}$.
7. $\{\varnothing\} \times \{0, \varnothing\} \times \{0, 1\}$.
8. $\{0, 1\}^4$.

Exercise 1.6. Sketch these Cartesian Products on the xy-plane (or in $\mathbb{R}^3$ for the last two).

(9) $\{1, 2, 3\} \times \{-1, 0, 1\}$	(15) $\{1\} \times [0, 1]$
(10) $\{-1, 0, 1\} \times \{1, 2, 3\}$	(16) $[0, 1] \times \{1\}$
(11) $[0, 1] \times [0, 1]$	(17) $\mathbb{N} \times \mathbb{Z}$
(12) $[-1, 1] \times [1, 2]$	(18) $\mathbb{Z}^2$
(13) $\{1, 1.5, 2\} \times [1, 2]$	(19) $[0, 1]^3$
(14) $[1, 2] \times \{1, 1.5, 2\}$	$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \times [0, 1]$

2 Solutions to Exercises From Section 1

Solution. 1.1

1. $\{5x - 1 : x \in \mathbb{Z}\} = \{\dots, -6, -1, 4, 9, \dots\}$
2. $\{3x + 2 : x \in \mathbb{Z}\} = \{\dots, -1, 2, 5, 8\}$.
3. $\{x \in \mathbb{Z} : -2 \leq x \leq 7\} = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7\}$.
4. $\{x \in \mathbb{N} : -2 < x \leq 7\} = \{0, 1, 2, 3, 4, 5, 6, 7\}$.
5. $\{x \in \mathbb{R} : x^2 = 3\} = \{-\sqrt{3}, \sqrt{3}\}$.
6. $\{x \in \mathbb{R} : x^2 = 9\} = \{-3, 3\}$.
7. $\{x \in \mathbb{R} : x^2 + 5x = -6\} = \{-2, -3\}$.
8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\} = \{-3, -2, 0\}$.
9. $\{x \in \mathbb{R} : \sin \pi x = 0\} = \mathbb{Z}$.
10. $\{x \in \mathbb{R} : \cos x = 1\} = \{2n\pi : n \in \mathbb{Z}\}$.
11. $\{x \in \mathbb{Z} : |x| < 5\} = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$.
12. $\{x \in \mathbb{Z} : |2x| < 5\} = \{-2, -1, 0, 1, 2\}$.
13. $\{x \in \mathbb{Z} : |6x| < 5\} = \{0\}$.
14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\} = \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$.
15. $\{5a + 2b : a, b \in \mathbb{Z}\} = \mathbb{Z}$.
16. $\{6a + 2b : a, b \in \mathbb{Z}\} = 2\mathbb{Z}$.

□

Solution. 1.2

17 $\{2, 4, 8, 16, 32, 64\} = \{2^x : x \in \mathbb{N}\}$

18 $\{0, 4, 16, 36, 64, 100, \dots\} = \{x^2 : x \in \mathbb{N}, 2x\}$.

19 $\{\dots, -6, -3, 0, 3, 6, 9, 12, 15, \dots\} = \{3x : x \in \mathbb{Z}\}$.

20 $\{\dots, -8, -3, 2, 7, 12, 17, \dots\} = \{5x : x \in \mathbb{Z}\}$.

□